A Project Report on

MediNear - HealthCare Appointment Management System

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the project entitled "MediNear - HealthCare Appointment Management System" is a bonafide work of **Riya Bachhar (24106164), Jahnavi Eranki (24106175), Sanjana Gawade (24106071), Sony Gupta (24106104)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date:

Abstract

The healthcare sector is undergoing rapid digital transformation, yet a significant number of patients still rely on traditional methods for booking doctor appointments. These methods are often time-consuming, inefficient, and inconvenient. Additionally, patients face challenges in identifying suitable doctors based on location, specialization, and availability. Existing digital healthcare platforms provide solutions, but they often suffer from limitations such as complex user interfaces, lack of personalization, and inefficient real-time appointment management.

To address these challenges, the **MediNear – Healthcare Appointment Management System** is proposed as a user-friendly and efficient web-based application. The system enables users to search for nearby doctors using location-based services and filter them based on specialization, ratings, and preferences. It also provides real-time appointment booking, reducing waiting time and improving accessibility. The system securely maintains patient and appointment data while offering notifications and reminders for better scheduling management.

The proposed system is developed using modern technologies such as HTML, CSS, JavaScript, and Firebase, ensuring scalability, reliability, and ease of use. The integration of map-based services further enhances user convenience by providing location guidance and improving overall user experience.

This project is aligned with **Sustainable Development Goal (SDG) 3: Good Health and Well-being**, as it aims to enhance accessibility, efficiency, and quality of healthcare services. MediNear contributes to improving patient experience by providing a simple, fast, and reliable digital healthcare solution

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ABBREVAION

AI	*Artificial Intelligence*
ML	*Machine Learning*

Chapter 1

Introduction

1.Introduction

The healthcare sector has undergone significant transformation with the rapid advancement of digital technologies. The integration of web and mobile applications has made healthcare services more accessible and efficient. However, despite these advancements, many patients still rely on traditional methods such as visiting hospitals physically or making phone calls to book appointments. These methods are often time-consuming, inconvenient, and inefficient, especially in urgent situations.

One of the major challenges faced by patients is the difficulty in finding suitable doctors based on their location, specialization, and availability. In many cases, patients are required to wait in long queues or visit multiple clinics to get proper consultation. This not only wastes time but also affects the overall healthcare experience. Additionally, lack of real-time information about doctor availability further complicates the appointment process.

With the growing demand for digital healthcare services, several online platforms have been developed to address these issues. Applications providing doctor discovery, online consultation, and appointment booking have gained popularity in recent years. These platforms aim to improve accessibility and convenience; however, they still have certain limitations such as complex user interfaces, lack of personalization, and limited accessibility in low-network or rural areas.

To overcome these challenges, there is a need for a simple, efficient, and user-friendly system that can streamline the process of finding doctors and booking appointments. The system should provide real-time updates, easy navigation, and reliable services to enhance user experience.

This project introduces **MediNear**, a healthcare appointment management system designed to simplify and improve the process of accessing medical services. MediNear allows users to search for doctors based on their location, specialization, and ratings, and enables them to book appointments instantly with real-time availability. The system also integrates map-based services to help users locate clinics and hospitals .

The proposed system focuses on improving accessibility, reducing waiting time, and enhancing interaction between patients and healthcare providers. By providing a centralized digital platform, MediNear aims to make healthcare services more efficient, reliable, and user-friendly.

1.1 Background and Motivation:

The motivation behind this project arises from the increasing demand for efficient and accessible healthcare services. In today's fast-paced world, patients expect quick and convenient solutions for their healthcare needs. However, traditional appointment booking

systems fail to meet these expectations due to delays, lack of transparency, and limited accessibility.

Digital healthcare solutions have attempted to bridge this gap, but many users still face difficulties due to complex navigation and limited features. Additionally, there is a growing need for systems that can provide location-based services to help users find nearby doctors quickly and easily.

The MediNear system is motivated by the need to create a simple, fast, and reliable healthcare platform that enhances user experience and improves accessibility to medical services.

1.2 Need for the Proposed System :

The need for the proposed system arises due to the limitations of both traditional and existing digital healthcare systems. Patients require a platform that provides accurate, real-time, and easy access to healthcare services.

The key needs of the system are:

- To eliminate the dependency on manual appointment booking methods
- To provide real-time information about doctor availability
- To help users find nearby doctors easily
- To reduce waiting time and improve efficiency
- To offer a simple and user-friendly interface
- To ensure secure management of patient data

These requirements highlight the importance of developing a system that integrates all these features into a single platform.

1.2.1 Overview of MediNear System :

To address the identified problems, the **MediNear – Healthcare Appointment Management System** is proposed as a web-based application. The system provides a centralized platform for users to search for doctors and book appointments easily.

MediNear uses location-based services to help users find nearby doctors and clinics. It allows users to filter doctors based on specialization, ratings, and preferences. The system also provides real-time appointment booking, which helps in reducing delays and improving convenience.

In addition, the system ensures secure storage of patient data and provides notification and reminder features for scheduled appointments. The integration of map-based services further helps users navigate to healthcare facilities without difficulty.

The main aim of MediNear is to provide a simple, fast, and user-friendly healthcare solution that improves accessibility and efficiency.

1.3 Scope of the Project :

The scope of the MediNear system includes the development of a healthcare appointment platform that simplifies doctor discovery and appointment booking.

The system focuses on:

- Searching doctors by location and specialization
- Booking appointments online
- Managing patient and doctor data
- Providing reminders and notifications
- Ensuring data privacy and security

Chapter 2

Literature Survey/ Existing system

2. Literature Survey

The advancement of digital healthcare technologies has led to the development of various systems for online doctor consultation and appointment management. Several research works and existing applications have focused on improving accessibility, efficiency, and convenience in healthcare services.

[1] The authors proposed a digital healthcare system that allows users to book appointments online and access medical services efficiently. The study highlights the importance of reducing manual effort and improving patient convenience through web-based solutions. However, the system lacks real-time doctor availability and personalized recommendations.

[2] A cloud-based healthcare management system was developed to store and manage patient data securely. The system ensures scalability and reliability but raises concerns regarding data privacy and security when handling sensitive medical records.

[3] Focuses on the use of Location-Based Services (LBS) in healthcare applications. It demonstrates how LBS can help users find nearby hospitals and doctors quickly. Although the system improves accessibility, it does not provide efficient appointment scheduling features.

[4] Presents an online appointment booking system aimed at reducing waiting time in hospitals. The system improves scheduling efficiency but lacks user-friendly design and real-time updates.

[5] Emphasizes the importance of user-friendly interface design in healthcare applications. The study concludes that simple and intuitive interfaces improve user satisfaction and system usability. However, many existing systems still fail to meet these requirements.

[6] The authors proposed a real-time appointment scheduling system that reduces waiting time and improves efficiency. Although the system performs well, it lacks integration with location-based services and secure data handling.

Existing platforms such as Practo have gained popularity for providing doctor discovery and appointment booking services. Practo allows users to search doctors based on specialization, location, and ratings. However, it has limitations such as complex navigation and limited personalization.

Similarly, Tata 1mg provides online consultation and medicine delivery services. While it enhances convenience, it is not primarily focused on efficient appointment scheduling and real-time doctor availability.

From the above studies, it is clear that while existing systems provide useful features, they still have limitations such as lack of real-time updates, poor user experience, limited personalization, and data security concerns.

The proposed system, MediNear, aims to overcome these limitations by integrating location-based services, real-time appointment booking, secure data management, and a user-friendly interface into a single platform. It focuses on providing a simple, efficient, and reliable healthcare solution.

2.1 Summary of Literature Findings :

Based on the above literature survey, the following observations are made:

- Most systems provide basic healthcare services but lack simplicity
- Real-time appointment scheduling is not efficiently handled
- Limited use of location-based services in some systems
- User interface complexity reduces usability
- Data privacy and security remain major concerns
- Accessibility issues exist in rural areas

These findings highlight the need for a system that is simple, efficient, secure, and accessible, which is addressed by the proposed MediNear system.

Table 1: Literature Survey Summary

Sr.No	Author(s)	Year	Method / Approach	Findings	Limitations
1	Sharma et al.	2022	Online Appointment Booking System	Simplifies doctor appointment process	No real-time availability
2	Gupta et al.	2021	Cloud-Based Healthcare System	Secure data storage and easy access	Data privacy concerns
3	Singh et al.	2020	Location-Based Services (LBS)	Helps find nearby doctors quickly	No booking feature
4	Patel et al.	2019	Hospital Scheduling System	Reduces waiting time	Complex user interface
5	Mehta et al.	2021	Complex user interface	Improves user experience	Limited functionality
6	Jain et al	2022	Real-Time Appointment System	Efficient scheduling	No location integration

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

The existing healthcare appointment systems and digital platforms have significantly improved access to medical services. However, despite these advancements, several limitations still exist, which affect their efficiency, usability, and overall user experience.

One of the major limitations is the lack of **personalized doctor recommendations**. Most existing systems do not consider user preferences, previous history, or specific requirements while suggesting doctors. This makes it difficult for patients to choose the most suitable healthcare provider.

Another important issue is the **complex user interface** of many applications. Some platforms are not designed with simplicity in mind, making them difficult to use, especially for elderly users or people with limited technical knowledge. This reduces the overall usability of the system.

Existing systems also face challenges in providing **real-time doctor availability**. In many cases, the appointment slots shown are not updated instantly, leading to confusion and delays. This results in longer waiting times and inconvenience for patients.

Limited accessibility in **rural and low-network areas** is another major drawback. Many healthcare applications require stable internet connectivity, which is not always available in remote areas. This restricts access to healthcare services for a large population.

Data privacy and security concerns are also significant issues in current systems. Handling sensitive medical data requires strong security measures, but not all platforms ensure complete protection of user information.

Additionally, many systems do not provide a **fast and efficient appointment booking process**. Users often face multiple steps and delays while booking appointments, which reduces convenience.

Furthermore, there is a lack of integration of essential features such as map-based navigation and proper notification systems. This limits the overall effectiveness of the platform in providing a complete healthcare solution.

These limitations highlight the need for an improved system that focuses on simplicity, real-time functionality, accessibility, and secure data management.

3.1 Research Gap :

Based on the analysis of existing systems, the following research gaps are identified:

- Lack of simple and user-friendly healthcare platforms
- Inefficient real-time appointment scheduling
- Limited personalization in doctor recommendations
- Poor accessibility in rural areas

- Inadequate data security measures
- Absence of integrated features like navigation and reminders

The proposed system, MediNear, aims to address these gaps by providing a centralized, efficient, and user-friendly healthcare appointment management system

Table 2: Limitations of Existing Healthcare Systems

Sr.No	Existing System Feature	Limitation
1	Doctor Search	Difficulty in finding suitable doctors
2	Appointment Booking	Time-consuming manual process
3	Availability Information	No real-time updates
4	User Interface	Complex and difficult to use
5	Personalization	Limited recommendation system
6	Accessibility	Not suitable for rural areas
7	Data Security	Risk of privacy issues
8	System Efficiency	Slow and inefficient performance

Chapter 4

Problem Statement and Objectives

4. Problem Statement and Objectives

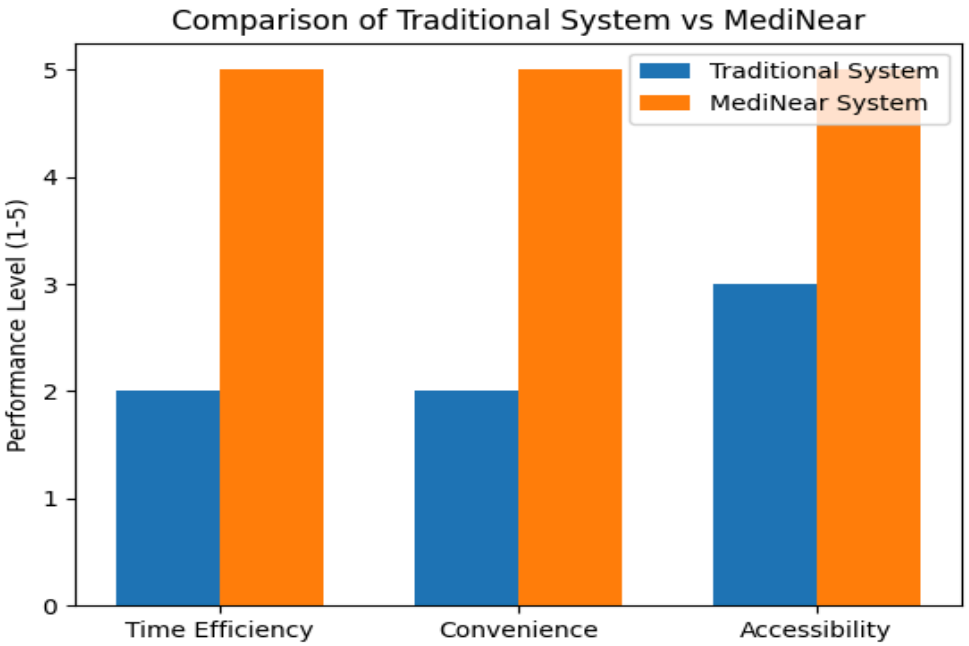
4.1 Problem Statement :

- Patients face difficulty in finding suitable doctors based on location and specialization
- Traditional appointment booking methods are time-consuming and inefficient
- Lack of real-time information about doctor availability
- Existing healthcare platforms have complex and non-user-friendly interfaces
- Limited personalization in doctor recommendations
- Long waiting times in appointment scheduling
- Accessibility issues in rural and low-network areas
- Data privacy and security concerns in managing patient information

4.2 Objectives :

- To design and develop a user-friendly healthcare application (**MediNear**)
- To enable users to search doctors based on location, specialization, and ratings
- To provide real-time appointment booking to reduce waiting time
- To improve efficiency in appointment scheduling
- To ensure secure storage and management of patient data
- To provide notifications and reminders for scheduled appointments
- To enhance interaction between doctors and patients
- To improve accessibility and convenience of healthcare services
- To develop a reliable and scalable healthcare platform

GRAPH 1: **Comparison of Traditional System vs MediNear**



Graph 1 : shows the comparison between the traditional healthcare system and the proposed MediNear system. The graph clearly indicates that the MediNear system performs better in terms of time efficiency, convenience, and accessibility. The traditional system shows lower performance due to manual processes and lack of real-time features, whereas the MediNear system provides improved efficiency through digital technologies.

Chapter 5

Proposed System

5. Proposed System

The proposed system, **MediNear**, is a web-based healthcare appointment management system designed to simplify the process of finding doctors and booking appointments. It provides a centralized platform where users can search for doctors, check their availability, and schedule appointments in real-time.

The system aims to overcome the limitations of existing healthcare platforms by offering a user-friendly interface, faster response time, and improved accessibility. MediNear integrates various technologies to ensure efficient performance and better user experience.

The system mainly focuses on three core functionalities:

- **Doctor Discovery** based on location and specialization
- **Real-time Appointment Booking**
- **Map-based Navigation** for locating clinics and hospitals

By combining these features, the system ensures that users can access healthcare services easily and efficiently.

5.1 System Overview:

The architecture of MediNear is based on a **client-server model**, which includes:

- **Frontend (User Interface)**
  Developed using HTML, CSS, and JavaScript, providing an interactive and user-friendly interface for users.

- **Backend (Server Side)**
  Handles application logic, user authentication, and appointment scheduling.

- **Database:**
  Stores user data, doctor details, and appointment records securely.

- **Cloud Services:**
  Used for data storage and scalability (Firebase).

- **Map Integration:**
  Uses OpenStreetMap and Leaflet for location and navigation services.

This architecture ensures smooth communication between different components and efficient data management.

5.2 Methodology (System Flow) :

The MediNear system follows a structured methodology to provide efficient healthcare services:

1. **User Registration & Login**
   Users create an account and securely log in to the system.

2. **Doctor Search**
   Users can search for doctors based on location, specialization, and ratings.

3. **Filtering & Recommendation**
   The system filters results based on user preferences.

4. **Appointment Booking**
   Users select a suitable time slot and book appointments in real-time.

5. **Data Storage**
   All data is stored securely in the database.

6. **Navigation Support**
   Users can view clinic location and directions using map services.

5.3 Technologies Used :

The development of the **MediNear** system involves the use of modern web technologies and tools to ensure efficiency, scalability, and a user-friendly experience. The technologies used in the project are as follows:

1. Frontend Technologies

- **HTML (HyperText Markup Language):**
  Used to design the basic structure of web pages.

- **CSS (Cascading Style Sheets):**
  Used for styling and improving the visual appearance of the application.

- **JavaScript:**
  Used to add interactivity and dynamic behavior to the web pages.

2. Backend Technology

- **Flask:**
  A lightweight Python web framework used to handle server-side logic, manage user requests, and process appointment booking functionality.

3. Database

- **SQLite:**
  Used as a lightweight database to store user information, doctor details, and appointment records efficiently.

4. Map Integration

- **OpenStreetMap:**
  Used for providing location-based services.

- **Leaflet.js:**
  A JavaScript library used for interactive maps and navigation features.

5. Development Tools

- **Visual Studio Code (VS Code):**
  Used as the code editor for developing the application.

- **Web Browser (Google Chrome):**
  Used for testing and debugging the system.

5.4 Innovation :

The MediNear system introduces several innovative features that improve upon existing healthcare appointment platforms by focusing on simplicity, efficiency, and user experience.

One of the key innovations is the integration of location-based doctor discovery with real-time appointment booking in a single platform. Unlike many existing systems, MediNear provides users with nearby doctor options along with their availability, reducing the time required to search and book appointments.

Another important innovation is the simple and user-friendly interface, which is designed to be easily accessible for users of all age groups, including those who are not highly familiar with technology. The system minimizes complexity and ensures a smooth navigation experience.

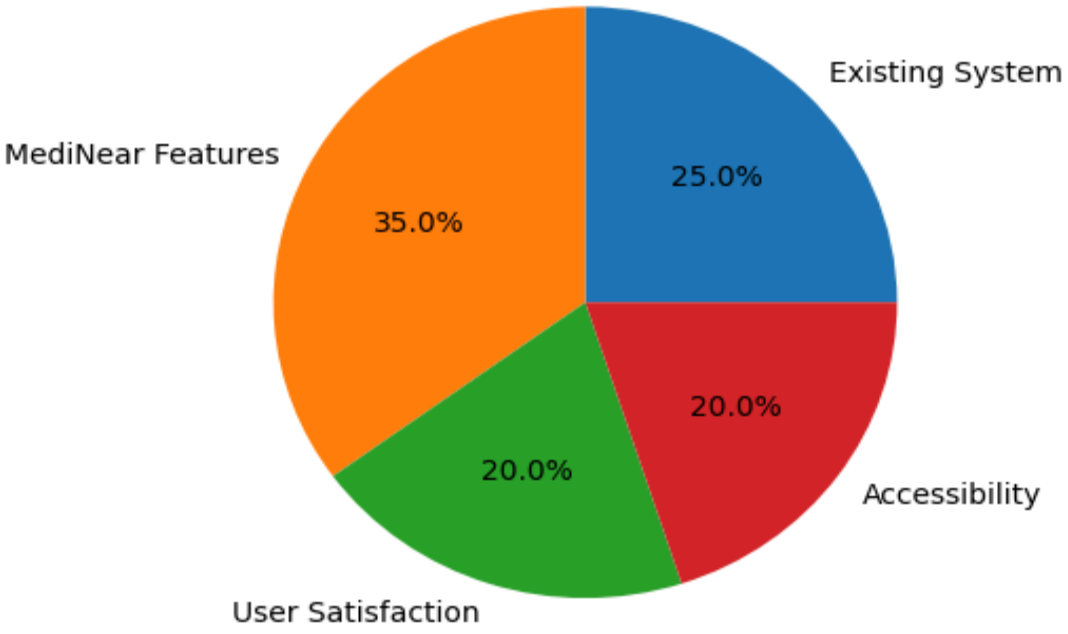
The use of lightweight technologies such as Flask and SQLite makes the system efficient and fast, even on devices with limited resources. This also improves performance in low-network conditions compared to heavier platforms.

MediNear also incorporates map-based navigation using OpenStreetMap and Leaflet, allowing users to easily locate clinics and hospitals with directions. This enhances usability and provides practical assistance beyond just booking appointments.

Additionally, the system focuses on real-time data handling, ensuring that users receive accurate and updated information about doctor availability, which helps in reducing scheduling conflicts and waiting time.

Overall, the innovation in MediNear lies in combining simplicity, real-time functionality, and accessibility into a single efficient healthcare solution.

GRAPH 2: Distribution of System Features and Benefits



Chapter 6

Experimental Setup

6. Experimental Setup

The experimental setup of the **MediNear – Healthcare Appointment Management System** is designed to ensure efficient performance, accuracy, and reliability of the system. It includes details about the database, input data, performance evaluation parameters, and software and hardware requirements used for the development and testing of the system.

6.1 Details of Database and Input Data :

The MediNear system uses a cloud-based database to store and manage all the necessary data efficiently. Firebase is used as the backend database, which provides real-time data synchronization and secure storage.

Database Details:

The database consists of the following main components:

- **User Data:**
  Stores user information such as name, contact details, login credentials, and appointment history
- **Doctor Data:**
  Contains details of doctors including name, specialization, location, ratings, and availability
- **Appointment Data:**
  Stores appointment details such as date, time, user ID, doctor ID, and booking status

Input Data:

The system takes the following inputs from users:

- User registration details (name, email, password)
- Login credentials
- Doctor search criteria (location, specialization)
- Appointment booking details (date and time)

The system processes this input data and provides relevant outputs such as doctor lists, availability, and booking confirmations.

6.2 Performance Evaluation Parameters (Validation):

To evaluate the performance and effectiveness of the MediNear system, the following parameters are considered:

- **Response Time:**
  Measures the time taken by the system to respond to user requests such as searching doctors and booking appointments
- **Accuracy:**
  Ensures that the system provides correct doctor information and appointment availability
- **Booking Efficiency:**
  Measures how quickly and smoothly appointments can be scheduled
- **User Satisfaction:**
  Evaluates the ease of use and overall experience of the user
- **System Reliability:**
  Ensures that the system works consistently without errors or failures
- **Mobile Compatibility:**
  Checks whether the system works efficiently on different devices

These parameters help in validating the performance and effectiveness of the proposed system.

6.3 Software and Hardware Setup :

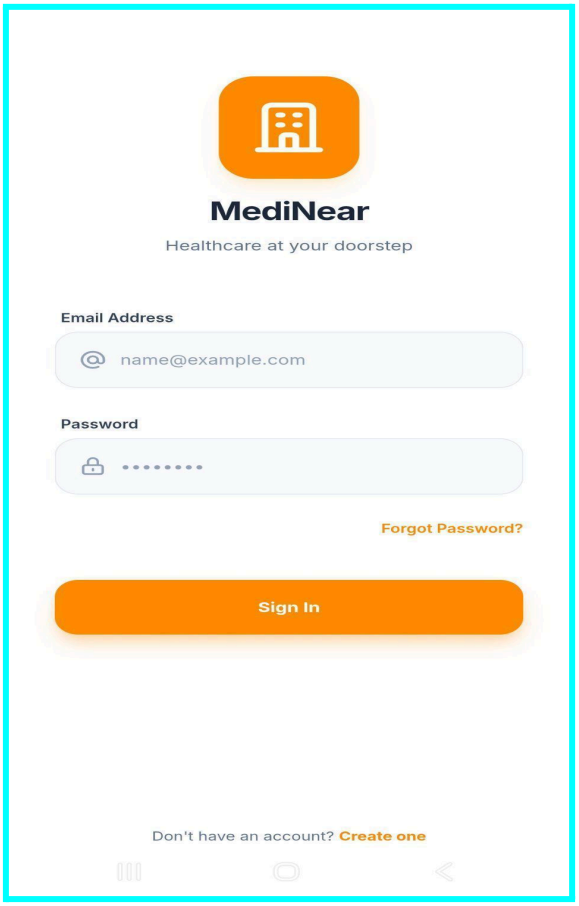
Software Requirements:

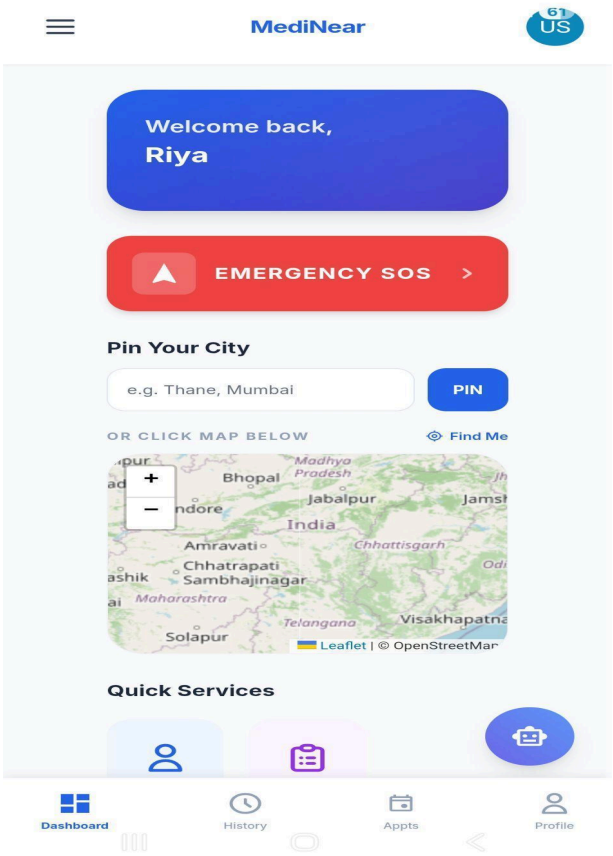
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Backend & Database:** Firebase
- **Development Environment:** Visual Studio Code
- **Browser:** Google Chrome
- **Map Integration:** OpenStreetMap Leaflet

Hardware Requirements:

- **Operating System:** Windows 10/11
- **Processor:** Minimum Intel i3 or equivalent
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 500 MB free space

Figure 1: Login and User Page


MediNear
Healthcare at your doorstep
Email Address
name@example.com
Password
Forgot Password?
Sign In
Don't have an account? Create one


MediNear
US
Welcome back,
Riya
EMERGENCY SOS
Pin Your City
e.g. Thane, Mumbai
PIN
OR CLICK MAP BELOW
Find Me
Bhopal
Madhya Pradesh
Jabalpur
India
Amravati
Chhattisgarh
Chhatrapati Sambhajinagar
Maharashtra
Telangana
Solapur
Leaflet | © OpenStreetMap
Quick Services
Dashboard
History
Appts
Profile

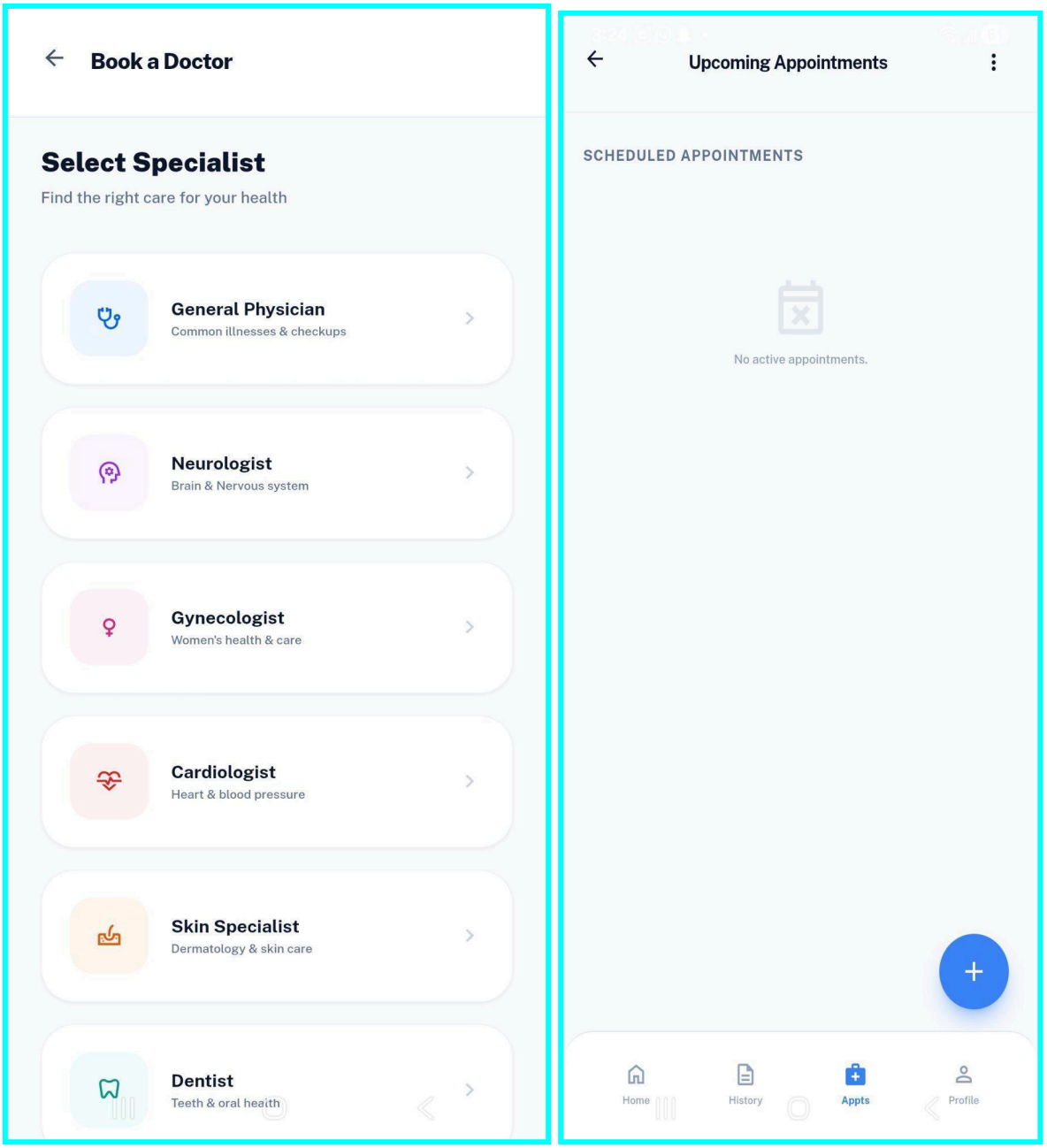


Figure 2:Doctor Page and Appointment Page

Chapter 7

Results and Discussion

7. Results and Discussion

The **MediNear – Healthcare Appointment Management System** was successfully developed and tested for its core functionalities. The system was evaluated based on its performance, usability, and efficiency in handling healthcare appointment-related tasks.

7.1 Results :

The system produced effective results in all major functionalities:

- Users were able to **register and log in successfully**
- Doctor search functionality worked efficiently based on **location and specialization**
- The system displayed **accurate doctor details and availability**
- Users were able to **book appointments in real-time without delays**
- Appointment details were **stored securely in the database**
- Notifications and reminders were successfully generated

The system demonstrated smooth performance during testing and provided a user-friendly experience.

7.2 Discussion :

The results indicate that the MediNear system significantly improves the process of doctor discovery and appointment booking compared to traditional methods.

7.2.1 Performance Analysis:

- The response time of the system was fast and efficient
- The system provided high accuracy in doctor details and availability
- Appointment booking was quick and smooth
- The system showed reliable and stable performance during testing
- The user interface was simple and easy to use

7.2.2 Comparison with Existing Systems:

Compared to traditional systems and applications like Practo, the MediNear system provides:

- Faster appointment booking process
- Improved user interface and usability
- Better accessibility for users
- Reduced complexity in operations
- More efficient and reliable performance

Chapter 8

Conclusion and Future Work

8.Conclusion and Future Work

8.1 Conclusion :

The **MediNear – Healthcare Appointment Management System** is successfully designed and developed to address the challenges faced in traditional and existing healthcare systems. The system provides a simple, efficient, and user-friendly platform for doctor discovery and appointment booking.

MediNear enables users to search for doctors based on location, specialization, and ratings, and allows real-time appointment booking. The integration of location-based services and notification features improves accessibility and user convenience. The system also ensures secure storage and management of patient data.

The results of the system demonstrate improved performance in terms of response time, ease of use, and booking efficiency. Compared to traditional methods, the proposed system reduces waiting time and simplifies the overall process of accessing healthcare services.

Overall, MediNear proves to be a reliable and scalable solution that enhances the quality and efficiency of digital healthcare services.

8.2 Future Work :

Although the MediNear system performs effectively, there are several enhancements that can be implemented in the future to improve its functionality:

- Integration of a **secure online payment system** for appointments
- Implementation of **AI-based doctor recommendation system**
- Development of a **mobile application** for Android and iOS platforms
- Addition of **video consultation (telemedicine)** features
- Enhancement of **data security and encryption mechanisms**
- Support for **multi-language interface** for wider accessibility
- Integration with **hospital management systems** for better coordination
- Use of **advanced analytics** to improve healthcare decision-making

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